

CHAMBERS'S JOURNAL

OF

POPULAR

LITERATURE, SCIENCE, AND ART.

Fourth Series

CONDUCTED BY WILLIAM AND ROBERT CHAMBERS.

No. 740.

SATURDAY, MARCH 2, 1878.

PRICE 1½d.

THE GAELIC NUISANCE.

SECOND ARTICLE.

A FEW months ago, in an article entitled 'The Gaelic Nuisance,' we endeavoured to point out the impolicy of fostering Gaelic as the vernacular tongue in the Highlands and Islands. Our observations were variously received. Many approved of the article; by some it was apparently misunderstood. On this latter account, we return to the subject, in the hope of removing such misapprehensions as may happen to exist. This time, at anyrate, we shall take care to be perfectly explicit as to our meaning.

In the article referred to, we offered no objection to the use of Gaelic, provided the young were brought up with a knowledge of English. That was distinctly our contention, and we believe that such is the opinion of all who think seriously on this important question. We therefore repeat in terms on which nothing but perversity can put a wrong construction, that the fostering of Gaelic to the exclusion of English—for it practically comes to that—is a grave error; it is a cruelty which merits exposure and reprobation. Why it is a cruelty is very clear. As previously stated, the use of Gaelic as the only known vernacular, keeps large numbers of poor people ignorant, it usually fixes them to their place of birth, and accordingly excludes them from earning their bread in the general competition of the world. It is very easy for enthusiasts living at a respectful distance to write in glowing terms about the antiquity of Gaelic, about the wonderful beauty of Gaelic poetry, about the philological value of Gaelic phraseology, about the satisfaction of being able to speak Gaelic as well as English. These are not the points in dispute. Let people, if they will, and if they can afford the expense, learn to speak and read Gaelic supplementary to English, just as many of us learn to speak and read French or German. The more languages that can be acquired the better. About that there is no contention. What we deem to be a

scandal and a cruelty is the practice of rearing, or allowing to grow up, groups of children with a knowledge of no other language than Gaelic; the consequence being that they are for the most part condemned to life-long poverty and ignorance. And that is what is done through the mistaken policy of it may be well-meaning sentimentalists and philanthropists, who are seemingly unaware of the misery they are helping to perpetuate. The English language, like the laws and constitution of the country, is a common heritage, in which every child has a claim to be instructed, so that all may be qualified to perform such duties as fall to their lot. Is it not, then, shocking to find groups of old and young scattered about the Highlands and Islands who cannot speak a word of English, and who cannot so much as sign their names? We might almost say they have no more knowledge of newspapers, or of English literature generally, than the lower animals, amidst which in dreary solitudes they hopelessly pass their existence.

The Highlanders have scarcely had justice done to them. They possess characteristics of a noble race. Faithful, honest, and steady in civil life. Valorous as soldiers. Peaceful and law-abiding in a very extraordinary degree. Those among them who by some good fortune quit their native glens and mix with the Lowland population, speedily learn English, and are able to converse as fluently in that language as in their native Gaelic. In fact, wherever they are brought in contact with English-speaking neighbours, they manifest no mental deficiency. In many instances they have attained to eminence. Only where they are habitually neglected, and left in untoward circumstances to vegetate in primitive ignorance, do they shew anything like laziness, and an indifference to improvement. From all we happen to know of the Highlanders, they only need to be put in the way of being cultivated by education and contact with the outer world.

In hinting at educational deficiencies we tread on tender ground. There is an Educational Act applicable to the whole of Scotland, whether the mainland or islands. No spot is exempted from

CHAMBERS'S JOURNAL

OF
POPULAR

LITERATURE, SCIENCE, AND ART.

Fourth Series

CONDUCTED BY WILLIAM AND ROBERT CHAMBERS.

No. 740.

SATURDAY, MARCH 2, 1878.

PRICE 1½d.

THE GAELIC NUISANCE.

SECOND ARTICLE.

A FEW months ago, in an article entitled 'The Gaelic Nuisance,' we endeavoured to point out the impolicy of fostering Gaelic as the vernacular tongue in the Highlands and Islands. Our observations were variously received. Many approved of the article; by some it was apparently misunderstood. On this latter account, we return to the subject, in the hope of removing such misapprehensions as may happen to exist. This time, at anyrate, we shall take care to be perfectly explicit as to our meaning.

In the article referred to, we offered no objection to the use of Gaelic, provided the young were brought up with a knowledge of English. That was distinctly our contention, and we believe that such is the opinion of all who think seriously on this important question. We therefore repeat in terms on which nothing but perversity can put a wrong construction, that the fostering of Gaelic to the exclusion of English—for it practically comes to that—is a grave error; it is a cruelty which merits exposure and reprobation. Why it is a cruelty is very clear. As previously stated, the use of Gaelic as the only known vernacular, keeps large numbers of poor people ignorant, it usually fixes them to their place of birth, and accordingly excludes them from earning their bread in the general competition of the world. It is very easy for enthusiasts living at a respectful distance to write in glowing terms about the antiquity of Gaelic, about the wonderful beauty of Gaelic poetry, about the philological value of Gaelic phraseology, about the satisfaction of being able to speak Gaelic as well as English. These are not the points in dispute. Let people, if they will, and if they can afford the expense, learn to speak and read Gaelic supplementary to English, just as many of us learn to speak and read French or German. The more languages that can be acquired the better. About that there is no contention. What we deem to be a

scandal and a cruelty is the practice of rearing, or allowing to grow up, groups of children with a knowledge of no other language than Gaelic; the consequence being that they are for the most part condemned to life-long poverty and ignorance. And that is what is done through the mistaken policy of it may be well-meaning sentimentalists and philanthropists, who are seemingly unaware of the misery they are helping to perpetuate. The English language, like the laws and constitution of the country, is a common heritage, in which every child has a claim to be instructed, so that all may be qualified to perform such duties as fall to their lot. Is it not, then, shocking to find groups of old and young scattered about the Highlands and Islands who cannot speak a word of English, and who cannot so much as sign their names? We might almost say they have no more knowledge of newspapers, or of English literature generally, than the lower animals, amidst which in dreary solitudes they hopelessly pass their existence.

The Highlanders have scarcely had justice done to them. They possess characteristics of a noble race. Faithful, honest, and steady in civil life. Valorous as soldiers. Peaceful and law-abiding in a very extraordinary degree. Those among them who by some good fortune quit their native glens and mix with the Lowland population, speedily learn English, and are able to converse as fluently in that language as in their native Gaelic. In fact, wherever they are brought in contact with English-speaking neighbours, they manifest no mental deficiency. In many instances they have attained to eminence. Only where they are habitually neglected, and left in untoward circumstances to vegetate in primitive ignorance, do they shew anything like laziness, and an indifference to improvement. From all we happen to know of the Highlanders, they only need to be put in the way of being cultivated by education and contact with the outer world.

In hinting at educational deficiencies we tread on tender ground. There is an Educational Act applicable to the whole of Scotland, whether the mainland or islands. No spot is exempted from

The Project Gutenberg eBook of Chambers's Journal of Popular Literature, Science, and Art, No. 740, March 2, 1878

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Chambers's Journal of Popular Literature, Science, and Art, No. 740, March 2, 1878

Author: Various

Editor: Robert Chambers
William Chambers

Release date: July 2, 2020 [eBook #62545]
Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/62545

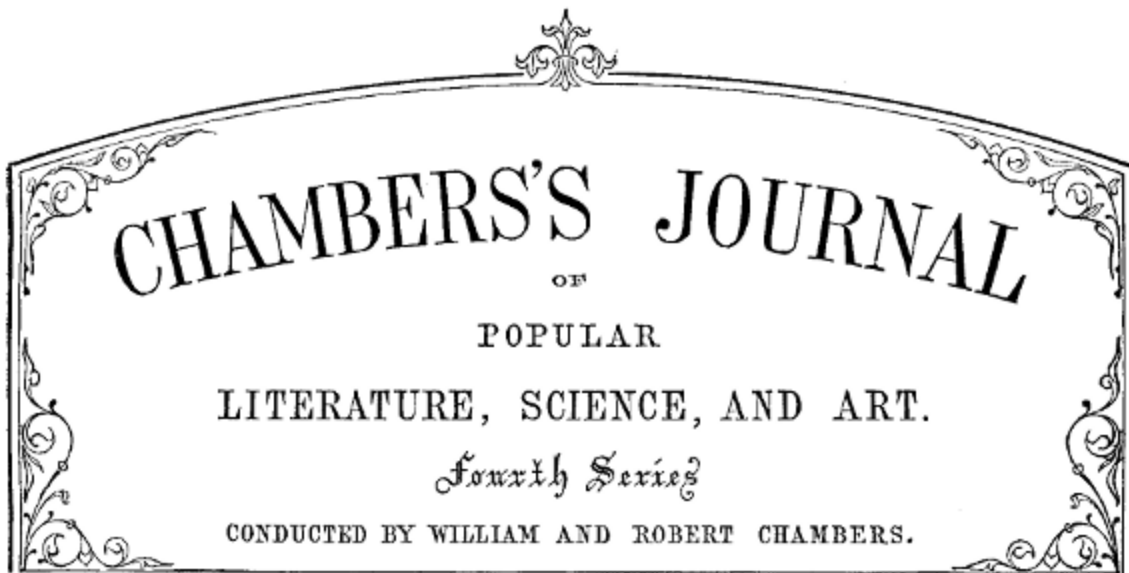
Credits: Produced by Susan Skinner and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
JOURNAL OF POPULAR LITERATURE, SCIENCE, AND ART, NO.
740, MARCH 2, 1878 ***

**CHAMBERS'S JOURNAL
OF
POPULAR
LITERATURE, SCIENCE, AND
ART.**

CONTENTS

THE GAELIC NUISANCE.
HELENA, LADY HARROGATE.
OUR SEA AND SALMON FISHERIES.
RACHEL LINDSAY.
IRRIGATION IN SOUTH AFRICA.
ALBATROSS NOTES.
A NOVEL LIFE-BOAT.
AWA'.



No. 740.

SATURDAY, MARCH 2, 1878.

PRICE 1½d.

THE GAELIC NUISANCE.

SECOND ARTICLE.

A FEW months ago, in an article entitled ‘The Gaelic Nuisance,’ we endeavoured to point out the impolicy of fostering Gaelic as the vernacular tongue in the Highlands and Islands. Our observations were variously received. Many approved of the article; by some it was apparently misunderstood. On this latter account, we return to the subject, in the hope of removing such misapprehensions as may happen to exist. This time, at anyrate, we shall take care to be perfectly explicit as to our meaning.

In the article referred to, we offered no objection to the use of Gaelic, provided the young were brought up with a knowledge of English. That was distinctly our contention, and we believe that such is the opinion of all who think seriously on this important question. We therefore repeat in terms on which nothing but perversity can put a wrong construction, that the fostering of Gaelic to the exclusion of English—for it practically comes to that—is a grave error; it is a cruelty which merits exposure and reprobation. Why it is a cruelty is very clear. As previously stated, the use of Gaelic as the only known vernacular, keeps large numbers of poor people ignorant, it usually fixes them to their place of birth, and accordingly excludes them from earning their bread in the general competition of the world. It is very easy for enthusiasts living at a respectful distance to write in glowing terms about the antiquity of Gaelic, about the wonderful beauty of Gaelic poetry, about the philological value of Gaelic phraseology, about the satisfaction of being able to speak Gaelic as well as English. These are not the points in dispute. Let people, if they will, and if they can afford the expense, learn to speak and read Gaelic supplementary to English, just as many of us learn to speak and read French or German. The more languages that can be acquired the better. About that there is no contention. What we deem to be a scandal and a cruelty is the practice of rearing, or allowing to grow up, groups of children with a knowledge of no other language than Gaelic; the consequence being that they are for the most part condemned to life-long poverty and ignorance. And that is what is done through the mistaken policy

of it may be well-meaning sentimentalists and philanthropists, who are seemingly unaware of the misery they are helping to perpetuate. The English language, like the laws and constitution of the country, is a common heritage, in which every child has a claim to be instructed, so that all may be qualified to perform such duties as fall to their lot. Is it not, then, shocking to find groups of old and young scattered about the Highlands and Islands who cannot speak a word of English, and who cannot so much as sign their names? We might almost say they have no more knowledge of newspapers, or of English literature generally, than the lower animals, amidst which in dreary solitudes they hopelessly pass their existence.

The Highlanders have scarcely had justice done to them. They possess characteristics of a noble race. Faithful, honest, and steady in civil life. Valorous as soldiers. Peaceful and law-abiding in a very extraordinary degree. Those among them who by some good fortune quit their native glens and mix with the Lowland population, speedily learn English, and are able to converse as fluently in that language as in their native Gaelic. In fact, wherever they are brought in contact with English-speaking neighbours, they manifest no mental deficiency. In many instances they have attained to eminence. Only where they are habitually neglected, and left in untoward circumstances to vegetate in primitive ignorance, do they shew anything like laziness, and an indifference to improvement. From all we happen to know of the Highlanders, they only need to be put in the way of being cultivated by education and contact with the outer world.

In hinting at educational deficiencies we tread on tender ground. There is an Educational Act applicable to the whole of Scotland, whether the mainland or islands. No spot is exempted from the operation of a school-board. Although the Act was passed in 1872, it appears from one cause or other that there are districts where no schooling is available, and children are suffered to run about wild. In an article in the *Scotsman* newspaper of January 5, 1878, a correspondent writing on the wretched condition of the Highland 'crofters,' or occupants of small patches of land, refers to the educational deficiencies in the parish of Barvas, on the west coast of Lewis. Here is what he says: 'At present, the children know not a syllable of English; the women and thirty per cent. of the men are as ignorant; and twenty per cent. of the people married cannot sign their marriage papers. One thing certain is that the people are themselves totally unprepared for

the good that the Act is expected to do them; and that it will be only by means of vigilant compulsory officers that its full operation will be secured. In the meantime the schools in the parish have not been opened; and ragged boys and girls hang about on the moor all day long herding cattle, or idle near the wayside in companies of threes and fours, holding fast by tethers, at the ends of which small melancholy lambs are grazing.' What a picture of primitive rural life! Education practically non-existent. The compulsory provisions of the School Act in a state of abeyance!

The island of St Kilda, to which we called attention, exhibits a small population with no means of learning English, and who for religious instruction in Gaelic are wholly dependent on the Rev. John M'Kay, a minister appointed by the Free Church. This worthy individual, who is a bachelor of advanced age, and whom, by mistake, we spoke of as being married, can speak and read English; but with the exception of the imported wife of one of the natives, he is the only individual on the island who can do so, and acts as a general interpreter on the occasion of visits from strangers. There is no school in the island, nor is there any attempt to teach English. Is this a condition of things which commends itself to philanthropists?

In a handsomely printed and illustrated work, *St Kilda Past and Present*, by George Seton, Advocate (Blackwood and Sons), 1878, there is an effective reference to the want of education in the island of St Kilda. 'Probably,' says this observant writer, 'the most beneficial influence that could be brought to bear upon the St Kildans would be of an educational kind. Through the instrumentality of the Harris school-board or otherwise, an energetic effort ought to be made to introduce a systematic course of instruction in English, with the view of the inhabitants enjoying the vast benefits which would inevitably ensue. At present, they are not only cut off from regular communication with the mainland, but in consequence of their ignorance of the language of the United Kingdom, they are debarred from the means of enlarging their minds, and subverting their prejudices, by the perusal of English literature. A recent number of *Chambers's Journal*—to which every English-speaking section of the globe owes such deep obligations—contains an admirable article, from the pen of the veteran senior editor, on the subject of "The Gaelic Nuisance," to which I venture to call the attention of all who are interested in the future welfare of the inhabitants of

St Kilda. The writer points to Galloway on the one hand, and to the Orkney and Shetland Islands on the other, as illustrative examples of the blessings which have flowed from the substitution of English for Gaelic and Norse respectively; and in the course of his remarks he makes special allusion to St Kilda.'

Thanking Mr Seton for this acknowledgment of the correctness of our views, we pass on to a note lately received from a sheriff-substitute in a Highland county. He says: 'Allow me to thank you for your article in the last part of *Chambers's* entitled "The Gaelic Nuisance." I have resided here for several years, and am convinced that the civilisation of the Highlands is impossible so long as Gaelic continues to be the language of the common people. I hope your article will open the eyes of common-sense people to the necessity of abolishing Gaelic as a spoken language, by the substitution of English.'

A gentleman connected by heritage with one of the outer Hebrides, sends us a note, in which, after commenting on the grotesque objections that had been made to our article, he observes: 'We all understand now—though a few may deceive themselves and others—that man is not made for language, but language for man. We Highlanders are determined to adopt the current language, just as we have adopted the current coin of the realm.' This is plain speaking; and we hope that the writer, using the power which his position gives him, will in his own locality see that the children are taught to read and understand English; such, in our opinion, whatever others may think, being only a simple act of justice.

In our former article we alluded to the case of Wales, in which large numbers are as unhappily excluded from a knowledge of the English language as are many of the Gaelic-speaking population of the Highlands. We are glad to see that this deficiency is beginning to attract attention, for reasons similar to those we employ. Recently at a large meeting in connection with the Welsh Church in Chester, presided over by the Bishop of Chester, as reported in *The Times*, Jan. 10, the Dean of Bangor, in speaking of Wales, remarked: 'Wales was in a certain extent backward. In the power of influencing those outside their own country, they were behind England, Scotland, and Ireland, simply because their language excluded them from making their thoughts and views known to those of different nationality.... He ventured to hope that the day was rapidly approaching

when every Welshman would be able to use the English language.’ Such a public acknowledgment as this is eminently satisfactory. It shews moral courage in combating popular prejudice. We should like to see Highland proprietors quite as openly avowing that it was time every Gaelic-speaking child ‘was able to use the English language.’

The most conclusive evidence that could be advanced respecting the serious disadvantage of maintaining Gaelic as an exclusively common language is that offered by Mr Simon S. Laurie, the accomplished Professor of Education in the University of Edinburgh, who lately delivered an Address on the subject of Education in the Highlands. According to a newspaper report of his address, he said in reference to the Highlanders: ‘One thing needful was to secure for them freedom of locomotion; so that when the pressure on one district became too great, the people might move to another. Without a knowledge of the English language, the country of the Highlander was bound round him as with a brazen wall. He need not try to get out of it, because his native language put him at such a disadvantage with other men that he had no chance against them.... There was no doubt that the teaching of Gaelic should be subordinate to the teaching of English. If they trained a boy in a Highland school to read, write, and speak Gaelic, what were they to do with him? How would we like to be in that position ourselves? Fancy a boy at the age of fifteen or sixteen able only to point out in Gaelic to a stranger the way he should take; would they not find that he had been miseducated—in fact cut off from being a member of the British Empire altogether? At the same time, while he held that, he was of opinion that they could not teach English to the Highlanders well except through the Gaelic. The Highland children learned very quickly—more quickly than the Lowland children—they could soon read with perfect fluency such a book as M’Culloch’s *Course of Reading*, and yet not understand a single word; shewing that they would not learn English well except through Gaelic. The aim of the whole teaching should be to make the pupils thoroughly acquainted with English.’

With such a concurrence of evidence, and with the knowledge that there is a School Act of six years’ standing, why, it will be asked, are children in the Highlands and Islands still left to remain untaught in the elements of education? That is a question that could perhaps best be answered by the Education Board for Scotland. We can only conjecture that the educational

deficiency in various quarters is due to the difficulty, for pecuniary reasons, in establishing and maintaining schools on a proper footing consistently with the obligations of the statute. Mr Laurie mentions that the school-rates press with a severity which in some places is perfectly paralysing. 'In Shetland, for example, the School Boards were brought to a stand-still. They could not face a rate of four shillings a pound; the same proprietors having to pay not less than four shillings a pound for poor-rate and other burdens besides.' This agrees with what we have privately heard of Shetland, where the rates of one kind or other very nearly swallow up the whole rental drawn by proprietors. Mr Laurie states emphatically as to this difficulty of school-rates, that 'unless the government paid what was necessary above fifteen-pence per pound, the Highlands and Islands would not have the full benefit of the Act of 1872.'

Evidently, the School Boards, notwithstanding their comprehensive and compulsory powers, are unable to plant and sustain schools in all quarters where required. The difficulty, it is observed, is financial. Let us instance the island of St Kilda. Its inhabitants are said to be seventy-six in number, while the annual rent exigible by the proprietor is somewhere about a hundred pounds, payable in kind. How can the School Board of Harris, with which the island is connected parochially, be expected to build a school and sustain a schoolmaster for the benefit of so small a population, in which there are perhaps only a very few children of school age? To build a school of the ordinary authorised type would cost at least six hundred pounds. And the payment of a teacher with other expenses would amount to one hundred pounds a year. The organisation of a school on this footing would go far beyond what is desirable or what could be asked for from either the state or the ratepayers.

A consideration of the financial difficulty leads to the conviction that something very much less costly than the present school organisation must in many parts of the Highlands be attempted, if the children are to get any education at all. Mr Laurie very properly remarks that children 'would not well learn English except through the Gaelic;' meaning by this, we suppose, that the teacher would require through the agency of Gaelic to explain the meaning of English words. That surely would not be difficult to accomplish; nor would it be unreasonable to establish schools on a much more modest footing than those latterly sanctioned by School Boards. The

Scotch were long accustomed to see a very humble class of schools in secluded rural districts. Often, these schools consisted of cottages of not more than two apartments, one of which constituted the dwelling of the teacher. These cottage schools were conducted at an exceedingly small expense, yet they answered their purpose. Neither dignified nor imposing, they were useful. They imparted to the few children in their respective neighbourhoods a knowledge of letters. We are inclined to think that a modification of this kind would solve some existing difficulties as concerns the establishing of schools among the sparse population of the Highlands and Islands. In short, it would be well to legalise a minor or sub-class of schools, to be conducted at a small cost, designed to effect a particular purpose, namely, that of communicating a knowledge of the English language to large numbers of poor children who are at present growing up in ignorance of any spoken tongue but their native Gaelic, and who, in many cases, as is seen, have no education whatever.

We hope the nature of our pleading is no longer misunderstood. It is, that all Gaelic-speaking children may in some shape or other be taught to read and understand the language common to the United Kingdom. There may be some statutory obstacles in the way. There should be none in the light of humanity and common-sense. Perhaps we may return to the subject. Considering that the welfare of successive generations of helpless beings is concerned, the subject is too momentous to be lightly treated, or to be swept aside by casual gusts of delirious opposition.

W. C.

HELENA, LADY HARROGATE.

CHAPTER XIII.—FATHER AND SON.

SIR SYKES was a weak man, and there are few readier elements of mischief than that of a weak man in a strong place—meaning thereby a position where there is authority to be abused. Some of the world's worst tyrants have been emphatically weak, mere spiteful capricious children grown to man's estate, and indued respectively with all the powers of the purple, the royal jika, and the triple tiara. But then the mighty system which they, unworthy, swayed, resembled some gigantic engine put into motion by the idle touch of a truant urchin's hand, and crushing all resistance by the resistless force of its swaying levers and grinding wheels.

A Devonshire baronet, in common with baronets elsewhere, does expect to be to a certain extent the petty autocrat of his own fields and hamlets, to find that there are those who court the great man's smile and tremble at his frown, and to hold rule within strictly constitutional limits over the dwellers on his land and the inmates of his house. The melancholy which had become a part of Sir Sykes Denzil's inner nature, and the indolence which had gradually incrusting him, had prevented the lord of Carbery from asserting in practice the prerogatives which he knew to belong to him in theory. Thus he did not really administer patriarchal justice on his estate, as some hale landlords do. His bailiff decided which labourers should be employed, which dismissed, and what wages should be allotted to crow-boys and weeding-girls. The steward arranged as to the barns to be rebuilt, the repairs to be granted or refused, the rent of whose cottage was to be forgiven, or which arrears were to be sternly exacted. Poachers whom the head-keeper did not like, found Sir Sykes's vicarious wrath make the parish too hot to hold them, while luckier depredators wired hares unpunished.

The part of a *roi fainéant* suited better with Sir Sykes Denzil's languid habits than they did with his tolerably active mind. He was well aware that the lethargy of King Log is always supplemented by the not wholly disinterested activity of King Log's zealous ministers, and had formed

frequent resolutions as to taking into his own hands the reins of government and becoming in fact as well as in name the lord of the manor—of six manors indeed, of which Carbery was the chief. These resolutions had never been acted upon; but Sir Sykes had always been able to lay to his soul the flattering unction that it rested with him alone to choose the time for realising them.

The events of the last few weeks had given some rude shocks to the baronet's indolent self-complacency. He had been threatened with consequences of which he, and he alone, could thoroughly comprehend the direful nature, and he had been forced to a series of compliances, each of which had degraded him in his own eyes. He had borne with the cynic effrontery of the sailor Hold. He had beneath his roof, seated at his table, in constant association with him and his, an unbidden guest. Mr Wilkins he had, through an unlucky chance, encountered, and instantly the fetters of a new vassalage appeared to be fastened on his reluctant limbs. And he owed this fresh humiliation to the misconduct of his own son!

Sir Sykes was very angry as he quitted No. 11 to seek out the chamber in which Jasper lay, so angry that his temper overmastered for the moment both the pleadings of natural affection and the instinct of caution. He laid his hand brusquely on the door of the room which had been pointed out to him as that to which Jasper had been conveyed, and was about to enter, with small regard to the nerves of the invalid within, when he felt a grasp upon his sleeve, and turned to be confronted by the wiry figure, anxious face, and bead-like black eyes of little Dr Aulfus.

'Excuse me. Sir Sykes Denzil, unless I am very much mistaken?' said the doctor, taking off his hat with such an air, that Sir Sykes, irritable as he was, felt compelled to acknowledge the bow. 'Allow me to introduce myself: Dr Aulfus, Benjamin Aulfus, Ph.D., M.D., M.R.C.S. of Heidelberg, Edinburgh, and London respectively. We never chanced, before to-day, Sir Sykes, to come personally into contact, and I regret that the occasion of our first interview should be so sad a one.'

During this speech, the doctor's eyes had taken stock as it were of Sir Sykes's aspect, and had read the signs of anger in his knitted brows and quivering mouth as accurately as a mountain shepherd discerns the portents of the coming storm. Nor was the reason far to seek. Gossip had been busy,

of course, with the private affairs of so exalted a family as that which dwelt at Carbery Chase; and Sir Sykes would have been astonished to hear at how many minor tea-tables the surgeon—for, his medical diplomas notwithstanding, Dr Aulfus was consulted nineteen times out of twenty as a general practitioner—had listened while Captain Denzil's debts and his father's displeasure were freely canvassed.

Of the arrival of Mr Wilkins and of the acceptances which the lawyer held, the little man of healing could of course know nothing. But he shrewdly surmised that Jasper had staked all that he could scrape together, and probably more, on the event of the desperate race which he had ridden on that day, and that his pecuniary losses had provoked the indignation of Sir Sykes, already smarting under recent sacrifices.

'You are very good, sir; I shall see my son, and then'——

Sir Sykes had got thus far in his speech, attempting the while to brush past the doctor, when he found himself gently but resolutely repulsed.

'Now, Sir Sykes,' said the little man, interposing his diminutive person between the tall baronet and the door, as some faithful dog might have done, 'pray have patience with me. Captain Denzil is my patient. He has sustained severe injury, the precise extent of which it is impossible yet for science to determine, and I am responsible for his safety, humanly speaking—the pilot, in fact, with whom it rests to bring him into port. We have just succeeded, by the help of an opiate, in inducing sleep. It will not last long, on account of the smallness of the dose. But it is of the utmost consequence that it should not be broken; and in fact, Sir Sykes, my patient is my patient, and I must protect him even against his own father.'

These last words were uttered in consequence of a renewed attempt on the baronet's part to force a passage, and the persuasive tone in which they were spoken contrasted oddly with the firmness of the doctor's attitude.

'Really, Mr Aulfus,' said Sir Sykes, half apologetically, half in dudgeon; but the other cut him short with: 'Excuse me, Sir Sykes. *Dr* Aulfus, if you please. It is perhaps the weakness of a professional purist, but I do like to be dubbed a doctor; as your noble neighbour and connection, the Earl, no doubt has a preference for the title of "My Lord." It has cost me dear enough, sir, that handle to my name; kept me, I may safely say, out of a

good four hundred a year of practice I might have had, since old women and heads of families are shy of sending for a regular physician; and that's why such fellows as Lancetter at High Tor, and Druggett the apothecary in Pebworth High Street, rattle about the county, feeling pulses and sending out physic, when a man who has more learning in his little finger than—— You smile, sir; and indeed I was unduly warm. No selfish love of lucre, believe me, prompted my remarks, but a sincere scorn for the prejudices and gullibility, if the word be not too strong, of our Devonshire Bæotians.'

By this time the doctor had succeeded in getting Sir Sykes into a neighbouring room, the door of which stood invitingly open, and thus securing the sleeper against the chance of being rudely awakened from his slumber. The baronet too had employed a minute or two in reflections which shewed him how unseemly was the part which he had been about to play, while some dim consciousness that it was unfair to visit on Jasper the unwelcome recognition and jocular impertinence of Mr Wilkins, began to creep into his perturbed mind.

'You forget, Dr Aulfus,' he said mildly enough, 'that I have as yet heard no details as to the injuries which my son has sustained. They are not, I apprehend, of a very serious or indeed dangerous character?'

'Umph! Dislocation of right shoulder, now reduced, but attended with much pain; severe contusion on temple; some bad bruises, and complete prostration of nervous system from the stunning blow and violent concussion of spinal cord,' dryly rejoined the doctor, summing up the facts as though he had been a judge putting the pith of some case before a jury. 'These are all the results that I know of'—— And he paused, hesitating, so that Sir Sykes for the first time felt a genuine twinge of alarm.

'Have you any suspicion, doctor, that there is something worse than this?' he asked, drawing his breath more quickly.

'I don't know. I hope not,' returned Dr Aulfus thoughtfully. 'Our knowledge after all is but cramped and bounded. I remember once at sea (I was assistant-surgeon in the navy and also on board Green's Indiamen, before I graduated in medicine) seeing a look in the face of a young sailor who had fallen from the mizzen shrouds to the deck, very like what I saw, or fancied I saw, in Captain Denzil's face to-day. But that was a fall, compared with

which even the accidents of a steeplechase are trifles,' added the doctor more cheerfully, and with an evident wish to change the subject.

'It is a mad sport, taken as a form of excitement,' said Sir Sykes, his resentment beginning to turn itself towards the institution of steeplechasing; 'worse still, when mere greed actuates the performers, brutal curiosity the spectators.'

'I quite agree with you, Sir Sykes, quite,' chimed in the doctor, with a bird-like chirrup of acquiescence. 'The mob, my dear sir, whether in decent coats or in torn fustian, is animated by much the same spirit which caused the Roman amphitheatre to ring with applause as wild beasts and gladiators, pitted against one another in the arena, stained the sand with'——

Here Captain Prodgers came in on tiptoe to say that Jasper was awake and sensible; that he had twice asked if his father had not yet arrived; and that he, Prodgers, had volunteered to make inquiries, and hearing the sound of voices as he passed the half-closed door, had entered. 'You, Sir Sykes, I have had the pleasure of meeting once before—at Lord Bivalve's, in Grosvenor Place,' he said with a bow. 'Captain Prodgers of the Lancers,' he added, by way of an introduction. The baronet returned the bow stiffly. He had some recollection of Captain Jack's jolly face beaming across the Bivalve mahogany; but he felt anything but well disposed towards the owner of Norah Creina and the man who had led his son into the present scrape.

'A friend of my son's, I am aware,' said Sir Sykes half bitterly.

'And I am afraid, "Save me from my friends," is the saying just now uppermost in your mind, Sir Sykes,' returned Captain Prodgers. 'But I do assure you that, hard hit in the pocket as I have been in this precious business, I'd sooner have lost the double of my bets, than have seen that poor fellow knocked about as he has been. I'm no chicken, and sentiment don't come natural to me, but I give you my word that had the tumble turned out as bad as I feared it would when first I saw it, I should—never have forgiven, myself, that's all.' Having said which, Jack Prodgers mentioned to the doctor that he should be found when required in the coffee-room, and with another bow to Sir Sykes, withdrew. The baronet, guided by Dr Aulfus, entered the darkened room where Jasper lay.

‘Is that you, sir? I thought you would come,’ said the hurt man from the bed, stretching out his feeble hand, and as Sir Sykes took the thin fingers within his own grasp, his anger, smouldering yet, seemed for the moment to die away, chased by the crowd of early recollections that beset his memory. He could remember Jasper as a lisping child, a quick intelligent boy, unduly indulged and pampered it is true, but bold-faced and free-spoken at an age when many a youngster, far nobler in every quality of heart and head, is sheepish and tongue-tied. In those days father and mother had been proud and fond of the boy, and Jasper’s future prosperity had been no unimportant element in Sir Sykes’s schemes and day-dreams.

‘You do not feel much pain now?’ asked the baronet gently.

‘In my arm and head I do,’ said the patient, stirring uneasily.

The doctor, as he adjusted the pillows, smiled hopefully. ‘A very good sign that,’ he whispered to Sir Sykes; ‘better than I had hoped for, after the draught. I think we may pronounce all immediate cause for anxiety to be over.’

‘When can he be moved?’ asked Sir Sykes, in the same cautious tone.

‘To Carbery? I should say, if he goes on as well as he is doing now, to-morrow,’ replied Dr Aulfus. ‘I will write down some instructions, with which it will be well to comply, for it will be some few days at least before he can resume his former habits of life.’

‘What are you two conspiring about?’ demanded Jasper, with an invalid’s customary peevishness, from the bed. And then Sir Sykes had to resume his seat and to say a few soothing words.

‘You’ll soon be well, my boy,’ he said kindly; ‘and sooner back with us at Carbery, under your sisters’ good nursing. Dr Aulfus here will, I hope, contrive to come over and give us a call every day till you get your strength again.’

Dr Aulfus said that he should be delighted to attend his patient at Carbery Chase, and indeed he looked radiant as he said it. A physician is, after all, a man, and probably a parent, and little Dr Aulfus had a wife and was the happy donor of six hostages to fortune. He valued the privilege of professional admittance at Carbery very highly, less on account of the emoluments directly derived therefrom, than of the many small people who

would augur well of his skill, since beneath a baronet's roof he should prescribe for a baronet's heir.

The brief conversation between Sir Sykes and his son was rendered the less marked because of Jasper's habitual reticence, and of his father's unwillingness to touch on any topic that might prove painful. Thus the lawyer and his bills met with no mention, and the steeplechase would also have been passed over, had not Jasper himself said: 'I told Jack Prodgers I shouldn't go in for cross-country work again, except with the hounds in winter. No fear, sir, of my donning the silk jacket any more, after this sharp lesson of aching bones and empty pockets. Don't be angry, please, though, with poor old Jack. He meant all for the best, he did.'

Sir Sykes replied that he had already had the pleasure of shaking hands with Captain Prodgers, whom he had formerly met, it appeared, in London society. And soon afterwards, in compliance with an almost imperceptible motion of the doctor's head, he withdrew; and Captain Jack was recalled to keep watch, uncomplainingly, beside his friend's couch, while the patient dozed or talked in snatches.

'Smoke away, old man; it rather does me good than not,' Jasper had said, and the captain's cigar was seldom extinguished during his vigil.

'He'll do!' was the little doctor's cheery whisper as he paid his early morning visit to his charge. And soon after noon, Jasper, pale and tottering, and with his bruised arm in a sling, was helped into one of the Carbery carriages and propped with cushions; and under the tender escort of his two sisters, Lucy and Blanche Denzil, was slowly and heedfully conveyed home to Carbery Chase.

OUR SEA AND SALMON FISHERIES.

IN the department of fishing-industries the march of scientific inquiry has already borne good fruit. The influence of the weather, or more properly speaking of the variations of temperature, on the plentifulness or scarcity of our food-fishes, has grown in importance as an element in determining the success or failure of the herring-fishery, for example; and at more than one fishing-station thermometrical observations are daily made by the fishermen, and reported to the meteorological authorities, who in their turn deduce generalisations and laws from the observations thus recorded. Thus the teachings of the formerly despised 'science' are beginning to bear fruit, and to be openly and fully recognised; and in the future, the fisherman, as a result of the generalisations just alluded to, may be able to determine with tolerable accuracy, before setting sail for the fishing-grounds, the chances of a successful or unsuccessful day's labour. Add to this, that, with increased knowledge of the conditions of life, development, and general history of our food-fishes, wise legislation may provide for the protection of these fishes and for the determination of the proper periods for the exercise of the fisher's art, and it will be owned that the gains from a scientific investigation of the fishing-industries are simply incalculable.

For these reasons we have peculiar pleasure in noting the appearance of a small volume, under the title of *Sea Fisheries*, by E. W. H. Holdsworth, and *Salmon Fisheries*, by Archibald Young, Commissioner of Scotch Salmon Fisheries (London: E. Stanford. 1877). The work is produced under the joint authorship of two gentlemen long connected with this important branch of British industry. To Mr E. W. H. Holdsworth has been allotted the task of giving an account of the sea fisheries of Britain; whilst Mr Archibald Young, one of the Commissioners of Scotch Salmon Fisheries, has undertaken the task of giving an account of the interests connected with the capture of the king of fishes. Mr Holdsworth has to do with the salt water, Mr Young chiefly with the fresh.

Within the last sixty or seventy years, the herring fisheries of Scotland, chiefly prosecuted on the north-east coast, have risen to be a most important

national industry and source of wealth, the value of the catch in a good year amounting to between two and three millions sterling. Needing no cultivation, the sea yields an annual harvest almost incredible in amount. Of course much capital is embarked; but without the hardihood, the enterprise, and the daring risks encountered by the fishermen, all would be unavailing. It is seen by a late Report, that in the united fisheries of herrings, cod, and ling, in 1876, nearly fifteen thousand boats, decked and undecked, were engaged, the total value of which amounted to upwards of a million sterling. Ever on the outlook for what will advance the interests of the herring fishery, the capitalists engaged in the business have latterly added a fast-sailing steamer to the fleets of boats; by which means herrings caught at a considerable distance are transferred from the boats to the steamer, rapidly brought into port, and being there properly prepared, are despatched by railway to various parts of the United Kingdom.

Railways, by facilitating transit, have been immensely advantageous to all kinds of fisheries. It might now be said that by this ready means of transit the most inland towns in the country are now as well supplied with fresh fish as towns on the coast; in many cases better. Ice has also played an important part in the transmission of fish to distant places. Salmon being thus preserved till it reaches the market, arrives in the best condition, and is sent to table fresh as when caught. One has only to look at the quantities of beautiful salmon and other fish spread out on marble benches of the fishmongers in any of our larger towns, to see what railways and ice have done for this branch of industry.

Mr Holdsworth expresses strong regret that the prospects of the Irish fisheries are not by any means of a promising kind, as far as the cultivation of the art or industry is concerned. All authorities agree in regarding the coasts of Ireland in most instances as representing fishing-grounds in which stores of wealth lie unheeded and uncared for. This is a state of matters much to be deplored, for the sake of all parties concerned—fishermen, consumers, and the nation at large. Some years ago, when we were in Ireland, we heard it mentioned that much of the fish sold in Dublin was supplied by fishermen from the coast of Wales; and we likewise heard that large quantities of dried white-fish were introduced to Portrush by fishermen from Islay and other western isles of Scotland. Though it is stated that the famine of thirty years ago has had much to do with the depressed

state of the Irish fisheries, and that emigration has also affected them, we yet fail to see why, by a little enterprise, the still resident natives should not be able to beat both the Welsh and Scotch out of their own market.

As regards the salmon fisheries, Mr Young leads us into a region which is still in some particulars a field of debate and controversy. There are very few readers, it may be presumed, who are ignorant of the controversies, for instance, which have been carried on concerning the correct answer to the question, 'Are parr the young of salmon?'—a query which Mr Young, along with the great majority of naturalists, answers unhesitatingly in the affirmative. The natural history of the salmon forms the starting-point of all knowledge of the fish, and of the information necessary for determining the conditions under which it may be properly and successfully caught—the terms 'properly' and 'successfully' in this case being taken as including the best interests of the fish and its race, as well as the interests of its human captors. Briefly detailed, the life-history of a salmon may be said to begin with the ascent of the parent-fishes in autumn and early winter to the upper reaches of our rivers for the purpose of depositing their eggs. In each salmon-mother it has been calculated about nine hundred eggs exist for every pound of her weight, and these eggs she deposits in a trench, excavated by aid of the jaw, in the gravelly bed of the stream. Fertilised after being deposited, by the milt of the male parent, the latter covers the eggs with gravel by means of his fins—the tail-fin being, as far as can be ascertained, the chief agent and means in effecting this necessary action. Such eggs as escape the attack of enemies—and of these, in the shape of aquatic birds and of other fishes, the salmon-ova have more than enough—undergo development, and are hatched in from ninety to one hundred and thirty days.

It would be an interesting study were we to trace the stages through which the young fish becomes evolved from its simple germ, and the wondrous formation of tissues and organs out of the soft jelly-like matter of which the egg is primarily composed. But want of space forbids; and our readers must therefore fancy for themselves the process whereby the hidden artist Nature works through development, and at length shapes out the young salmon, or 'parr.' It may be mentioned in proof of the small proportion borne by the salmon-eggs actually deposited, to those developed, that authorities agree in stating that out of three thousand eggs deposited, scarcely *one* egg may

survive—so terrible is the destruction of young salmon. This fact alone, as Mr Young argues, should tell powerfully as an argument in favour of *artificial* propagation; since out of three thousand eggs which are thus hatched, at least one thousand young fishes may be successfully reared.

The curious fact is noticed that in most if not all broods of salmon, half of the parrs will become ‘smolts’—as they are called in their next stage—at the end of a year or so, whilst the other half will not become smolts *until after the lapse of two years and more*. This incongruity, if we may so term it, has led to the questions, ‘Do the parrs become smolts between thirteen and fifteen months after they have left the egg, or at the age of two years and two months?’ Both questions may apparently be answered in the affirmative, since each brood exhibits this peculiar feature of some of its members coming to the smolt-stage long before the others. Mr Young remarks on the authority of a salmon-breeder in the north, that about eight per cent. of the salmon hatched by this gentleman became smolts at the end of the first year; about sixty per cent. at the end of the second year; and about thirty-two per cent. at the end of the third year. These facts would seem to indicate that the end of the second year is the most natural period for the assumption of the smolt-guise, which, as distinguished from that of the parr, exhibits a beautiful coat of silvery mail.

The parr, it may be remarked, dies if placed in sea-water, whereas the smolt thrives in the latter element. On reaching the sea, the young smolt may measure from four to five inches. After a residence in the sea of some six or eight weeks, the smolt returns to its river as a ‘grilse,’ which varies from five to eight or nine pounds in weight, according to the time it has remained in the sea. After returning to its river the grilse spawns, and then returns to the sea. The features of the mature salmon are now apparent, and the fish increases in size after each such annual migration to the sea. Indeed nothing is more extraordinary in the history of the salmon than its rapid increase and growth after these periodical migrations to salt water. Three salmon which weighed ten, eleven and a half, and twelve and a half pounds as they were migrating seawards, were duly marked; and on being caught six months afterwards when returning to the fresh water, were found to have increased in weight to the extent of seventeen, eighteen, and nineteen pounds respectively. Although salmon usually return to the rivers in which they first saw the light, yet it has been ascertained that the practice is not an

invariable one. There is no good reason why one river should not suit a salmon as well as another, and in their wide migrations these fishes are exceedingly likely to enter rivers other than their native streams.

One of the most interesting topics touched upon by Mr Young in his observations, is that regarding the relative late or early development of salmon in different Scotch rivers. Prefacing, that a 'clean' salmon is a fish that has been for some time in the sea, it has been generally believed that rivers which issue from a lake are 'early' rivers—or in other words that they are streams which clean salmon will ascend in the early spring. But this idea receives little or no support from facts as they stand. Many early Scotch rivers have no lake heads; whilst many Scotch rivers which run out of or through lakes are late rivers. Mr Frank Buckland thinks a river's 'earliness' in the matter of salmon depends on its proportion of mileage in length to its square mileage of 'catchment'—that is of the land-area from which the river is fed. This, however, seems to us a whimsical theory, and might be disproved by facts. As regards the 'earliness' of rivers, Mr Young's theory is that much depends on temperature; in fact, temperature is known to be the chief cause which regulates the distribution of life in the sea, and there is no one fact, so far as we are aware, which can be said to militate against his views. His theory is, however, being tested by the Scottish Meteorological Society at Inverugie; by the Duke of Richmond, on the Spey; by the Duke of Sutherland, on two early and two late rivers in Sutherlandshire; and by the Tweed Commissioners—the method of testing being by thermometers applied to the fresh water of the rivers, and to the sea near their mouths.

The latter part of the volume under notice is occupied with statements relative to salmon fishery laws and legislation, a subject in which the author is naturally deeply interested, and in which our knowledge of the salmon naturally culminates when the fish is regarded from an economic standpoint. In Scotland, it seems we are far behind England and Ireland in respect that there are no Inspectors of salmon fisheries empowered to make annual inspections and reports on the Scotch salmon fisheries! And this fact becomes the more inexplicable, and the more urgently demands remedy, when we consider that the Scotch fisheries are many times as valuable as those of our English neighbours. Then also, Mr Young has a most justifiable grumble at the fact that, in our statutes, there are very inadequate provisions

made for the removal of artificial and natural obstructions in salmon rivers, and for the prevention of pollutions; and no close-time for trout or char. The importance of clearing away natural obstructions to the ascent of the salmon in rivers is well exemplified when it is found that in Scotland no less than 478 miles of river and loch are thus closed against these fishes. No less forcibly shewn is the vexatious fact that rivers are polluted and rendered unfitted for breeding-streams by means and methods which the River Pollution Commissioners in their Reports declare to be preventable at a moderate cost, without injury to the manufactures with which they are connected.

Besides pollution, two things are especially detrimental to the Scottish salmon fisheries. The first to be mentioned, is the abominable practice of building weirs across rivers in order to send water into mill-lades, and the ignoring of the law that requires that the water shall be periodically diverted into the river again. Certain proprietors, to make the most of their lands, give perpetual leases of ground to manufacturers of one kind or other, with liberty to build a weir and take water to turn their machinery. There may be provisions in the lease as there is in law to the effect that the withdrawal of water shall cease during the night and on Sundays. Such provisions are, however, in many instances neglected, as giving too much trouble. The result is, that the whole river, or very nearly the whole, except in times of flood, is diverted into the mill-lade, whereby trout and salmon are unable to surmount the weir, and are effectually barred from getting to the upper part of the stream. In plain terms, by the selfishness of a proprietor (or a pair of them, one on each side), all who dwell on the river above the weir are deprived of the fish which nature had bountifully assigned to them. Already in these pages we have alluded to a scandalous case of this kind on the Tweed.

The second of the two things which act detrimentally on the Scottish salmon fisheries is the circumstance that certain landed proprietors near the mouths of some rivers possess a right to establish nets for the purpose of catching all the salmon that attempt to go up the stream. We do not contest the legality of their arrangements. We only speak of the cruel way it acts on the rights of all who live in the upper parts of the river, and on whose waters the salmon have bred. While the lower proprietors catch the great bulk of the fish, those higher up get but a miserable remnant. During the whole of

the time that the nets are on, the lower proprietors have a practical monopoly of the fishings. Is that at all reasonable? As a consequence, first of the weirs, and second of the netting system of the lower proprietors, there is evoked throughout the upper part of rivers in Scotland, a gloomy and almost vengeful hatred of the existing salmon-fishing system. Of course the higher and middle classes take no part in demonstrating their sense of the injustice that is committed. The lower classes, less scrupulous, and indignant at the rapacity of the weir-owners and lower proprietors, take such salmon as they can get hold of in spawning-time, thus destroying by myriads in embryo what should have been a vast national advantage. Detesting as we do all sorts of poaching and irregularities, we are glad that the Commissioners appointed to investigate the condition of the Scottish salmon fisheries, have laid stress upon the miserable imperfections to which we have ventured to draw attention.

Mr Young informs us that in 1874, as many as 32,180 boxes of Scotch salmon were sent to the London market alone, the estimated value of which might possibly be L.321,800. It seems to us, however, a hard case that the great bulk of such valuable property should be secured by proprietors at or near the mouths of the several rivers, to the exclusion of those in the upper reaches of the streams, who ought to have an equal right to participate in the annual fish-harvest. Free-trade in salmon-fishing, so much as lies within the limits of strict justice, is still in expectation. We commend the subject to the further consideration of Frank Buckland, Mr Young, and brother-anglers.

RACHEL LINDSAY.

A SOUTH-AUSTRALIAN STORY.

TOWARDS the end of November, about two years ago, I received the following curt note from my brother Donald, who like myself is a sheep-farmer in South Australia. 'MY DEAR JERRY, Lizzie sends her love, and hopes to see you when your shearing is over, as usual. If you'll say what day, I'll fetch you from Ballarat.—Yours affectionately,

DON GARDINER.'

'N.B.—Just begun to wash the wool. Lizzie's sister says she has seen my apparatus at Conolly's, but I don't think it. Ask Conolly.'

Conolly was a neighbour of mine, and he chanced to have brought me Don's letter, and to be lighting his pipe at my elbow while I read it.

'Conolly,' said I, 'do you know any of Mrs Gardiner's sisters? She has an unlimited number, I believe, for I have met a fresh one—sometimes two fresh ones—every Christmas for about half-a-dozen years, and here is still another I never heard of. She appears to be acquainted with you and this neighbourhood'——

'O yes; that's Cinderella,' interrupted Conolly, as he abstracted a bundle of newspapers from our joint post-bag and began to rip the wrapping from them. 'Haven't you seen Cinderella? She was never out of Tasmania, I suppose, until last spring, when she was staying up here with the Macdonalds. The Macdonald girls called her Cinderella because she had always been the one to stay at home and keep house while the others went about. Her proper name is Rachel. O Jerry, Jerry!' he broke out suddenly, laughing in what seemed to me a very offensive manner (my proper name I may mention being Gerald), 'your sister-in-law Lizzie will be too many for you. She won't let you escape this time. She has kept Rachel as her last card.'

'If ever I marry a woman with such a name as that, I hope I shall be a henpecked husband for the rest of my life!' I retorted angrily, seizing a

paper-knife and beginning to tear away at the *Australasian*, so as to drown further conversation upon what was a very sore subject.

My brother Donald's wife Lizzie was as good and kind a little woman as ever breathed, but like many young wives in happy circumstances, she was a matchmaker. And being impulsive, effusive, and not quite—what shall I call it? I don't like to say she was not quite a lady, but that would suggest my meaning—she did not pursue her calling with that tact and judgment which its delicate nature required. I need not say more, except that she had a number of spinster sisters, and one only bachelor relative, who lived all by himself in single-blessedness on his own fine and thriving property, and that I was that male victim. I beg pardon of all the Misses Lindsay for using such a term; I was not a victim as far as *they* were concerned. But I did feel it hard that I should be laughed at wherever I went as the captive knight of half-a-dozen fair ones, when I had never had the choosing of one of them.

When I received the above letter I had just seen my last wool-bale packed on the last bullock-dray and started on its slow journey to Melbourne; and the day after I set off myself on my yearly visit to Don. He was less fortunate in respect of sheep-shearing than I, for living in an exceptionally cool district, where an exceptionally wet and wintry spring had kept everything behindhand, he had still all his troubles to come. I thought of that as I buttoned myself into my Ulster, which I was glad of that cold morning, though Christmas was only a month off; and I reflected that I should be the only unemployed man at the disposal of the household until the shearing was over, and foresaw (as I thought) the consequences. I made up my mind, however, that I would defy Lizzie's machinations in a more systematic manner than heretofore. May I be forgiven for so priggish a determination.

It was midnight before I reached Ballarat, where Don usually met me; but upon this occasion I found a telegram stating that he was too busy to leave his farm, and would send for me next day. So I had one game of pool at the club and went to bed; and next morning enjoyed an hour or two over newly arrived English papers and periodicals, and a solitary lunch; and then the familiar old ramshackle buggy and the beautiful horses Don was famed for made their appearance, and I set off on the last stage of my journey. When I arrived at my destination it was dark and raining heavily; and the groom who opened the stable-gate told me that my brother had not long come up

from the wash-place and was interviewing shearers at the hut. I was wet and muddy, so I went straight to my room without even asking for my sister-in-law, who was usually in her nursery at that hour, and proceeded to make myself respectable for dinner. Presently I heard Don about the passages (the house was 'weather-board' and the partitions extremely thin) asking the servants where I was; and then his head and a half-bared neck appeared in the narrow aperture between my door and the door-post.

'Glad to see you, old boy; but I'm too dirty to come in,' said he. 'Seen Lizzie?'

'Not yet.'

'Seen Rachel?'

'Not yet. But I say, old man, would you mind telling me how many *more* sisters you've got?'

'No more,' said Don with a grin. 'She's the last one, and she's the best of them all.'

'Then I hope I may be allowed for once to enjoy the society of one of Lizzie's sisters, in a friendly way,' I grumblingly responded (for I may as well admit that Don and I had had confidences of old on this subject). 'Don't you think you could give Lizzie a notion that I don't mean to get married, or that I've a sweetheart up the country, or something of that sort?'

'Not I,' rejoined my brother, laughing. 'I'm not going to spoil her fun, poor little soul; you're old enough to take care of yourself.' And with that he went off, whistling cheerfully, to his dressing-room.

When I had completed my toilet, I gathered up some boxes of choice cigars that I had been purchasing in town, and carried them to the door of the adjoining apartment, which had been Don's smoking-room ever since I had known it. To my surprise, the bolt shot sharply as I touched the handle, and I heard a rustle of drapery inside. A housemaid coming along with lamps for the dinner-table called out hastily: 'O sir, that is Miss Rachel's room now. The smoking-room is at the end of the verandah, where Miss Carry slept last year. Mrs Gardiner wished it to be changed because she didn't like the smell of tobacco so near the bedrooms.'

I took back my boxes, thinking no more about it, and went on to the drawing-room, which was full of light and warmth and comfort, as usual, and where I found two of my little nieces sitting demurely on a sofa in their best frocks, ready to rush into my arms. Lizzie came hurrying in after me, rosy and radiant, and with plenty of flounces and colours about her, and gave me her own enthusiastic welcome; and then Don, spruce and perfumed, joined us. Don in his early years had been a dandy, and a little youthful weakness remained in him still. He never came to dinner without rings on his fingers and subtle odours in his clothes; and he was at great pains to keep a pair of Dundreary whiskers accurately adjusted on each side of a closely shaven chin. He had been ten years in the Bush, and had never objected to 'roughing it' in a general way; but he persisted in shaving himself every morning, let what would, happen; which singular habit in an Australian country gentleman very much puzzled his bearded friends. I for one, used to quiz him as well as I knew how, when I saw him swathe his neck in a handkerchief, before going out to his work, if the sun shone too strongly; but I respect his little vanities nowadays, and hope he will keep his white throat and his Dundreary whiskers as long as he lives, bless him. He took one of his little girls on his knee, and questioned me about my station matters and about Conolly's sheep-wash (which was *not* so well furnished with improvements as his own, much to his satisfaction); and Lizzie gave me an account of the development of her respective children since I had seen them last, including the cutting of the baby's teeth; and then the dinner-bell rang.

'Where's Rachel?' inquired Don.

I turned a languid eye upon the door when we heard the sound of a distant rustle, expecting to see one of the smart and smiling damsels I was so used to, and wondering whether this one would be dark or fair. With a slow and quiet step she came along the hall and entered the room, and my heart began suddenly to beat in a very unpleasant manner. She had a delicate, thoughtful, but piquant face, wavy brown hair modestly and becomingly set, and a slight figure daintily dressed in pale blue silk, with a little white lace about throat and arms; and yet she was the image of Marie Antoinette in Delaroche's picture, only with a more majestic dignity of carriage, if that could be, and a more cold and calm disdain upon her face. As soon as I saw her, and felt the exceedingly faint acknowledgment she vouchsafed when

we were formally introduced, I intuitively guessed—with a horrible sense of shame and mortification—that she had overheard what I said to Don in my bedroom through those card-paper walls!

I never thought I should feel so concerned at standing ill with one of Lizzie's sisters as I felt before that evening came to an end. All through dinner I saw, without looking, offended dignity in the poise of her head and the studied repose of her manner, and heard the ring of it in every inflection of her voice, though it was so subtle and delicate that only a guilty conscience could detect it. It was a great deal worse in the evening, when Lizzie began her fussy little contrivances for throwing us together. The poor little woman never had so impracticable and aggravating a sister to manage; and I never met one who attempted to treat me with such open indifference and thinly-veiled contempt. It is unnecessary for me to state the consequences. I began to interest myself in this Miss Lindsay as I had never interested myself in the others. I began to hanker for her good opinion, as I had never hankered for theirs. I longed to set myself straight with her, and beg her forgiveness for a thoughtless insult that I would have given worlds to recall, and to feel that the way was open between us to meet and associate as friends. This longing grew apace as the evening wore on, but the prospect of its gratification grew less and less. Until the little ones were taken away by their nurse she devoted herself to them, telling them stories most of the time in a dark corner, whence merry chatter and ripples of subdued laughter came flowing out to us; but when they were gone, the bright vivacity that was her true characteristic went too, and she became Marie Antoinette again.

With an amiable wish to make things pleasant, Lizzie asked her to pour out the tea; but she merely stood in front of me at the tea-table, with her little nose in the air, and asked me whether I took sugar and cream, in a high clear tone that brought a puzzled wonder into her sister's face and a slow smile to Don's. I came and stood beside her, to take the cups from her hand (her pretty head was about level with the flower in my button-hole), and she tried to ignore me, but could not. Her hands shook slightly and a little angry flush came and went in her face; but I preferred that to having no notice at all. Later on she went to the piano, and sung song after song for the delectation of Lizzie and Don, neither of whom had the hearing ear and the understanding heart to appreciate the pure quality and poetic sweetness of

her voice. By this time I was very low-spirited, and I drew away from my host, who was growing sleepy after his hard day's work, and took a chair near her—which of course was a signal to Lizzie to leave the room. As she sung on, forgetful of me and of everything but the poetry awaking in her, and as I studied the pose of her slight figure and half-bent head, and the now dreamy happiness in her face, and listened for the first time after many years to the true translation of a language that I loved, a vague perception dawned in me that there was some latent fellowship between us. And then I felt that Fate had indeed been hard.

The silence of the room was presently brought into strong relief by a deep snore from Don, whereupon she suddenly rose from the piano and saw that we were virtually alone. 'Good-night, Mr Gardiner,' she said promptly, holding out a somewhat reluctant hand and stiffening back into her unnatural stateliness.

I took it and held it and looked into her face; and I tried to tell her, as well as plain 'good-night' would do it, that I knew what had happened and wanted her to forgive me. I think she guessed what my look meant, by the sudden crimson flame in her face; but she walked out of the room with as much dignity as she had first walked into it, without another word.

The early days of December were cold and wet, and the shearing was a protracted and troublesome affair. Don hovered about restlessly, whether in or out of the house, always bothered and anxious, and paying frequent visits to the barometer. The ladies clung to their fireside as if they had been in England; and I tied myself to Lizzie's apron-string with an abject alacrity that puzzled and charmed her. My opportunities for 'improving the occasion' were many, but somehow I could never turn them to account. The pride of that little maiden was quite beyond my management. Lizzie threw us together; she left us alone; she did all that in her lay to further my desires for a reconciliation and an understanding; but the implacable resentment of the last of the Lindsays towards me for that wretched slip of the tongue was a stone wall I could not climb over. The worst of it was, she did and said nothing tangibly offensive; and I was precluded by all sorts of considerations from mentioning the subject we were both (and that we both knew we were) thinking of. So matters went on day after day. And before a

great many days were past I was over head and ears in love—I may as well say it and have done with it—and began to feel desperate and dangerous. She walked about the house with her grand air, my Queen Marie Antoinette, my little tyrant that I could almost have demolished with a finger and thumb; and I, standing six feet three in my stockings, had to acknowledge that she was invincible as well as unmerciful. Unregenerate savage as I was, I had faint longings now and then to take her by those slender shoulders, and shake her.

There were times when she became her sweet self, and could not help it though she tried; and these times were born of music. She and I both loved music with that special love that nature permits to a few people; but to no one else in the house did the ‘heaven-born maid’ present attractions. Don, hard at work all day, could go to sleep after dinner in his arm-chair; and Lizzie, after her manner, could go out of the room in the middle of the most charming song. Then, when we were singing together, or when she was gentle and gracious with the spirit of melody in her, then was the oil thrown upon my troubled waters. At times such as these it flashed across me that she was aware of it.

At length on one of these occasions I made a dash at her guarded citadel; I will not say in what words, but with the blundering foolishness that I suppose characterises all implied aspirations; albeit with sufficient plainness to leave no chance of being misunderstood; and then I had indeed to bite the dust for once in my life. She had been singing *Ruth* with the most touching pathos and abandonment—‘Where thou goest I will go, and there will I be buried’—and I could not stand it. Don was in the room, but snoring in that comfortable undertone which denoted a sound and quiet slumber. She stood with her back to the piano, and the sheet of music trembling and rustling in her hands, watching his nodding head in the distance, and turning her delicate profile to my view.

‘No; I will not,’ she half whispered with haughty rapidity. ‘You should have known I would not. I do not particularly want to marry anybody,’ she added, flashing round upon me with her crimson face; ‘but I will never marry *you*. I made up my mind to that long ago.’

Everybody knows how, in the supremely solemn moments of one’s life, one is apt to be assailed with most incongruously ludicrous ideas. In spite of my

bitter mortification at her reply, an absurd rhyme that I had heard somewhere, flashed into my head:

Do not be like Nancy Baxter,
Who refused a man before he'd axed her.

I believe she saw the ghost of a smile that might have hovered round my eyes when I begged to know why she had made up her mind never to marry me; and that made her savage.

'Because you think I came here to be made love to,' she retorted, with all the concentrated contempt that her sweet face and voice could hold. 'You think Lizzie and I have been plotting to catch you—you think we wanted to inveigle you into marrying me! I know what you are going to say'—as I rose and seized her hands, to stop her—'but it is not the truth. I *heard* you'—lifting her angry eyes, now wet with unshed tears, to mine—'I heard you, with my own ears, tell Don to warn Lizzie beforehand that you did not want to be married.'

'I know you did,' I replied, tightening my hold of her hands, while she made feeble efforts to get away; 'and I wish my tongue had been cut out before I could have insulted you and her like that. Forgive me, Rachel; I have been punished enough.'

'I cannot,' she answered, still panting with her excitement. 'I should be ashamed of myself if I could take a man who had even *thought* of me like that.'

Two tears began to trickle from her eyes, and a little hysterical catch in her breath betrayed to me that her defiant courage was failing her. I would not let her go. Love and shame and a resentful disappointment made me a little savage too.

'I never *did* think of you like that,' I said sternly; 'and you know it. I must hold you till I clear myself—I cannot bear it'——

A log tumbled in the grate, and Don woke up. She caught away her hands and sped out of the room; and I walked through a French window into the cool summer night, too full of wrath and love to speak to anybody.

This was how we stood when at last (on Saturday, the 18th of December) the true Christmas weather came, and we found ourselves in the hot afternoon alone on the croquet-lawn—alone for the first time since my stormy wooing was interrupted. Don being still busy in the sheep-yards and shearing-shed, I had been playing singly against Lizzie and her; and now Lizzie had been called away to the nursery to consult with a needlewoman who was at work there. We were both anxious (though for very different reasons) to leave off playing when our chaperon had departed; but it was not easy to do so in the middle of a game, especially as she had instructed her partner to play for both of them until she returned. So we knocked the balls about for a few minutes in embarrassed silence, and then had an altercation as to which hoop Lizzie had been through; and then we both got a little huffy, and played, first with indifference, and then with a malicious energy, which resulted at last in my sending her partner's ball into the thickest Portugal laurels in the shrubbery.

‘Oh, I beg your pardon!’ I exclaimed with compunction, as she solemnly marched off to look for it. ‘Let me find it for you.’

‘Do not trouble yourself,’ she replied sharply; and immediately dashed in between the laurel and a very prickly rose-bush, whose long sprays caught her muslin dress and tore it. I saw her straw-hat amongst the big dark branches, and her little hand searching under them for a moment or two; then she started up suddenly with a quick cry, and bounded into the path again.

‘What is the matter? Have you hurt yourself,’ I asked anxiously.

Her hat fell to the ground, and she stood before me with the blazing sun on her pretty head, and a wide-eyed horror in her face. ‘Wait a minute for me!’ she panted breathlessly; ‘I want you to help me—I have been bitten.’ Before I could collect my senses to understand what she meant, she had sped like a flash of light into the house; and dashing into the laurel bush, I saw what had happened. A big black snake was gliding away from the spot where she had been kneeling.

What was to be done? I stood still for a moment paralysed; then I sent up a hurried prayer for help, and simultaneously ‘cooeyed’ three or four times with all the force of a powerful pair of lungs, for Don at the wool-shed. Then I hurried after her, and met her coming through the door of my

brother's dressing-room with one of his razors in her hand. Her face was white and set as she seized my hand and hurried me into the smoking-room, which was near us, and turned the key in the lock. I knew what she wanted; and I set my teeth in an agony that no words could express, and which I can never think of now without a shudder.

'Look!' she said piteously, with a little sob in her throat; and I looked, and saw one of the fingers of her left hand tied round tightly with a piece of string below the first joint, and the end of it already livid and swollen and shewing the unmistakable punctures of the snake's fangs. She laid her other hand on my arm, and looked up at me with a beseeching face that nearly unmanned me.

'Help me!' she whispered eagerly; 'now—now; before the others come!' And she held out the razor, open and shining. 'It is no use to suck it—it only wastes time,' as I seized her finger and put it in my mouth. 'Don't, don't! I want to be on the safe side. I don't want to die! O pray, pray help me!'—now sobbing passionately—'or else I must try to do it myself. I won't cry out; I won't mind it. I will turn my head away.' She laid her finger on the edge of the table, and I took the razor from her, and with all the courage I could muster, excised the wounded part. She bore the cruel operation without a murmur.

An hour afterwards the commotion in the house was over, but the shadow of death was on it. Rachel was in her bed, white and faint and breathing heavily, twitching with weak fingers at the bedclothes, and staring with dull eyes into the sad faces around her. I knelt in my room close by with my head on my outspread arms, weeping like a child as if my heart would break, and listening to the creaking of the doctor's boots and the whisking of skirts and whispering of awed voices on the other side of the thin wall. There was nothing else that I was privileged to do, now that I had done that dreadful thing which they told me might be the saving of her precious life.

As the twilight fell, the voices in the sick-room took a louder and more cheerful tone; and presently one of them called softly: 'Jerry, I want you.' Lizzie met me in the passage with a tremulous tear-stained smiling face. 'The doctor says she will be all right now, and that she has to thank you for

it,' she whispered. 'Don't stay here any longer; go and have a cigar with Don.'

I seized her hand and kissed it, and looked at her with my wet eyes full of foolish emotion, too glad for speech; and the brightening intelligence of her countenance was curious to note. 'I thought you didn't care for each other,' she said archly; 'but,' she added drily, 'I suppose I was mistaken.'

'Don't suppose anything, Lizzie, there's a good girl. But let me know when I may see her,' I replied earnestly.

'All right—I understand—I'll let you know,' she said, nodding her head vigorously with an air of mystery and importance; and then I went, not to have a cigar with Don, but to walk about the dark garden alleys, alone with my thoughts.

Our patient improved steadily all night, so much so that the family assembled at breakfast as usual. Then a great hunt was made for the snake (at Lizzie's instigation, on the children's behalf), which lasted a long while and was wholly unsuccessful. Then church-time came, and the buggy was ordered to take the servants and the little girls to church; and the hot day wore on. Towards evening, as I was loafing about the garden, Lizzie came running across the croquet lawn—where the balls and mallets were still lying about as we had left them, though it was Sunday—and told me that Rachel was up and dressed, and that she chanced to be alone in the drawing-room.

I stole in to her in the twilight with my heart beating fast; and for a few moments she did not notice me. She was standing by the open piano, small and white and weak, with a shawl wrapped round her, gazing at the silent key-board, with tears running down her face. No one could look less like Delaroche's Marie Antoinette than she looked then.

I took three long steps and reached her side; she gave a great start and turned round to meet me. 'I shall not again be able to play to you for a long while!' she said, looking up at me for sympathy in this new trouble with her soft wet eyes.

When she said that—instead of making me the little speech I had expected, thanking me for saving her life—I put out my arms. And though we said no word, we forgave one another.

And how pleased Lizzie was when she saw the last of the Lindsays transferred to my unworthy self.

IRRIGATION IN SOUTH AFRICA.

THE irrigation of lands by water-channels connected with rivers is accepted as an important means of agricultural development in countries subject to protracted droughts, where rain falls only at distant and uncertain intervals. The irrigation caused by the periodic overflow of the Nile is a noted case in point. But for the annual overflow, temporarily deluging the land for a foot or two, Lower Egypt would be barren instead of a scene of fertility. In a country like England, where there is, generally speaking, too much moisture, the chief consideration is to dry the land sufficiently by draining, instead of flooding it with water; the lesson being thus taught that as regards the culture of soils every country must act according to circumstances. In India, also in Ceylon, there are some remarkable instances of the value of irrigation, and in these countries much more of the same kind requires to be done to avert the horrors of lengthened drought and famine. On this subject, we propose to say a few words regarding large tracts in Southern Africa, which are very much in the condition of those parts of Egypt fertilised by the waters of the Nile.

We speak first of the river Oliphant, which falls into the sea on the west of Cape Colony, and which has various important affluents. The land through which these streams flow is of a most desolate character—broad belts of sand, interspersed with low scrubby bush, swelling into moderate hills, with rugged mountains for a background. Upon the country in the lower part of the Oliphant River rains have no appreciable effect; but when the soil is thoroughly soaked by the overflow of the streams, after the periodical inundations, and then covered by the deposit brought down by the floods from the upper districts, its fertility is wonderful. The average yield is more than a hundredfold. The quantity of land of this character along the Lower Oliphant alone was estimated by the government surveyor in 1859 as eight thousand seven hundred acres.

Thus, like Egypt with its Nile inundations, those districts of Cape Colony—otherwise almost barren—are annually fertilised. But unlike Egypt, the country is unprovided with any means for utilising to the full extent the

advantages thus conferred. No appliances are prepared for the purpose of storing the water thus brought down; no artificial channels are cut for directing it and spreading it over a large area; and when the short rainy season has passed, the inhabitants are content to sit down and wait for the next 'periodical.'

A characteristic story is told by a colonist who visited the locality some years ago: 'I strolled along the banks of the river, and was much struck with the extremely fertile appearance of the soil, and the very little which had been done for turning it to account. It seemed as if the Creator had done everything for the country, and man nothing. Scarcely any rain had fallen for some time past, and the river had not overflowed its banks for more than a year. The stocks of grain and vegetables were getting very low. The farmer was complaining much about the long protracted drought; and when he had finished, I took the liberty of pointing out how he could, by leading out the stream for the purposes of irrigation, or by fixing a pump to be propelled by wind, on the river's bank, secure an abundant supply independent of the weather. He seemed to listen with some interest to the development of my plans; and I began to hope that he had decided upon doing something to relieve himself of the difficulty; but eventually, after turning round and scrutinising the whole horizon in the direction of the river's source, as if in search of some favourable symptom, he yawned heavily, and merely observed in Dutch: "Oh, it will rain some day!"'

Of the Zout or Holle River, the most northerly of the tributaries of the Oliphant, Mr P. Fletcher, the government surveyor, says: 'By its arteries it brings together the rich karroo soil of the Hantam and Hardeveld and the rich sandy soil of Bushmanland. The best crop of oats I have seen in Africa was in the deposit of this "periodical." Other portions are of a very saline character. At a rough guess, I believe that in many spots a dam might be constructed three or four feet high, and a couple of hundred feet long, which would flood several hundred acres, thereby rendering them richly arable. I have measured some of last year's "slick" two feet deep; this, of course, was under the most favourable circumstances; but by the use of dams, the deposit might be regulated, the fresh slick might be allowed to deposit to its full extent, so that in a few years the lands would be out of the reach of ordinary floods, if desirable that they should be so. By this system of irrigation, even the most saline basin would become available to

agriculture, and about nine or ten thousand acres on the banks of this one periodical river might be brought under cultivation, which would even excel the richest soil in the “Boland” (upper country).

‘Several tributaries to the Zout River have extensive karroo deposits; some of their basins reaching to nearly one mile in breadth, and their fall being so little, that, standing in their delta, a person cannot sometimes judge with the eye which direction water would flow. Their water-course, which winds through the middle of the deposit, is always well defined, and shews a longitudinal section of the plain. Except in ordinary heavy rains, those channels carry off all the water without overflowing, while a few pounds would leave them in a condition to produce fifty, eighty, or even one hundred-fold. Such is the nature of several tributaries of the Hartebeeste River. I have not seen the latter, but have been more than once informed that it has in some places a deposit of five miles in breadth, that when it does overflow, there is abundance of grass for all the cattle that visit that quarter. If this description of the Hartebeeste River be correct, the products it may be able to yield either in the form of grain or pasture for cattle would appear to most people fabulous. We have here, and not here only, but over an extensive portion of the whole colony, the richest soil in the world lying at present for two-thirds of the year utterly unoccupied, waste and worthless.’

The Hartebeeste is the last principal tributary, from the south, of the Orange River, and rises in the same chain of hills as the Tanqua, one of the tributaries of the Oliphant—namely, the Roggeveld Berg, receiving affluent streams from the south-east and south-west, draining in fact nearly the whole of the central northern part of the colony. This district is at present almost neglected. The chain of hills in fact, which runs from east to west across the centre of the colony, cuts off the northern half from the mass of the colonists, notwithstanding the fact that here is the most fertile land in South Africa.

The Zak, as the upper part of the Hartebeeste is called, is another instance of the wonderful effects upon the soil of periodical inundations. The following particulars are given in Mr Noble’s book on *Cape Colony*. In the dry season these streams are comparatively small, and often a mere succession of pools; but after rains they run briskly, and where level with the banks, overflow and soak the adjacent flats. In many places so very

even is the country that they may be said to have no defined channel, and form extensive sheets of water a few inches deep. The Zak River at two hundred and fifty miles from its source thus varies in breadth from one to four miles; and further on from Onderste Doorns to Leeuwenkop it widens as much as ten miles. Along its course is the most valuable part of Great Bushmanland. Water can be obtained in its bed even when dry, and its valley generally affords pasture to cattle during both the winter and summer months. After floods, there are extensive alluvial bottoms on each side of it, where agricultural products of every kind might be raised. These are now commonly used by the squatters as sowing lands, without any labour or trouble beyond scratching in the seed. One overflowing of the soil is sufficient to insure a crop even although no rain should fall afterwards. The returns are something marvellous, especially those of wheat.

In 1859 the number of Europeans settled on the irrigable portion of the Oliphant was estimated at one hundred and twenty souls. When there was an overflow of the river, they were active enough; day and night they worked incessantly; the sun and the moon alike witnessed that they did not eat the bread of idleness. But talk to them of improvements in the way of artificial irrigation by dams or pumps, and they ridiculed the idea. Such was the description given at that time; and such, with very little alteration, is an accurate statement of affairs now. The population of Calvinia and Clanwilliam, the two districts drained by the Oliphant River, was in 1875, 15,856, of whom only 2046 were classed as 'urban.' These figures of course include the natives as well as the settlers; but they represent an enormous advance in population since 1859. It is probable that if measures were taken to secure the permanence of the advantages which are now only temporarily enjoyed, the population and wealth of the districts would rapidly increase.

Of the fertility of the soil without any attempt at cultivation, there are abundant evidences. A sandy plain apparently as barren as the Sahara itself is suddenly transformed into an expanse of waving grass for hundreds of miles, so soon as the annual rains occur. This 'twaa-grass' or Bushman grass is an excellent fodder for horses and cattle, which thrive and grow fat upon it in a few weeks: even when dried up in the winter it is better feeding than any available green pasture. The natives scratch in their seeds and leave them to ripen, which they do without the least attention, and whether the country is visited by drought after the summer rains or not.

The principal drawback to the complete cultivation of the lands is the absence of roads or water-carriage. The Orange River, though a magnificent stream, and navigable in certain parts of its course, is blocked by narrow gorges, shallows, falls, and other impediments, and is useless as a water-way. In time probably, it may be made available, by means of inter-communicating canals to enable the rapids, &c. to be passed; but at present the community must look to the extension of roads and railways for the means of fully utilising the produce which would be raised if a ready market could be found. In this respect a decided step has already been taken. A railway of ninety miles in length has been constructed from Port Nolloth on the north-west coast, in Namaqualand, to the upper mining districts, its terminus being at Ookiep. On the most difficult portion of the route the cost has been very little more than a thousand pounds a mile for this distance; and it might be extended further eastward at a still less cost, to the great advantage of the country drained by the Hartebeeste. Another railway, or a good system of roadways, is wanted to open up the Oliphant water-shed; and with these means of carrying away the produce—all that is necessary—the immense natural resources of the district would be fully developed. In the Fraserburgh district, where the Upper Zak river rises, substantial houses, springs, wells, and dams have already been constructed, and plantations and gardens are being extended; but then from Fraserburgh excellent roads run east and west; and the railway to Cape Town comes as far as Beaufort, situated sixty or seventy miles south-east.

Finally, in Great Bushmanland, diamonds have been found; but there are far greater and more permanent sources of wealth than diamond-fields. Sheep and oxen can be raised, and their wool and hides turned to profitable account. Wheat, grapes, and vegetables of all kinds will grow in abundance. In fact, for pastoral and agricultural qualities the country is unsurpassed. Here then is a field for the enterprising emigrant from our own country. Capital alone is wanting for its development; and capital however small, judiciously expended, must be at once remunerative. We are glad to be able to add that an Act has been passed by the Cape Parliament for granting facilities to landowners for obtaining by loan or otherwise the means of improving their lands by irrigation or other similar permanent works.

Other districts prove how the colonists have succeeded in turning what was, more really than Great Bushmanland or the Lower Oliphant can be said to

be, a 'howling desert' into valuable farms, by opening up springs, making dams, forming irrigation channels, and planting trees where no trees existed, and where water was only an occasional and very ephemeral visitor. There is no reason indeed why the 'Nile lands of South Africa' should not rival in productiveness the great 'world's wonder' in the north of the continent; after which, from natural circumstances, they have been not inaptly named.

ALBATROSS NOTES.

FAR out in the Atlantic and Pacific Oceans, often two thousand miles and more from the nearest land, sails the albatross in its graceful and powerful flight; now following in the wake of the ship, to catch any chance morsel that may have fallen from the cook's waste-basket; now skimming along the water, and occasionally snapping up some small ocean-waif from the crest of a wave; or with a few vigorous strokes of its broad wings, gliding easily round and round the vessel, though she may be going at the rate of a dozen knots an hour.

No passenger to southern lands can have failed to note the extraordinary powers of flight of this magnificent bird, and the wonderful ease with which it sweeps for some minutes together through the air on expanded motionless pinions, rising and falling slightly, and taking advantage of the gravity of its own body and the angle at which the wind strikes its feathered sails to prolong the course of its flight with the least possible effort. Seldom, except in very calm weather, may it be seen to alight upon the water, from which it rises with difficulty, running for some distance along the surface. The ends of the wings clear of the water, it turns towards the breeze, and rises into the air in a gentle curve, in exactly the same manner as a paper kite. That the albatross follows a ship for many days in succession, sleeping at night upon the water, and coming up with her in the morning, there can be no doubt. We have watched them for several consecutive evenings during fine weather, in the latitude of the trade-winds, settling down on the water at sundown, and preening their feathers, until they became mere specks in the field of the telescope; but they were with us again in the morning soon after sunrise; some strangers among them perhaps, but several which, from some peculiarity of marking, we knew to be our companions of the day before. In one instance, a conspicuous mark had been made by a pistol-bullet in the wing of an old brown-headed and curiously pied bird, by which he could be identified beyond doubt. The second or third flight-feather had been shot away, leaving a clearly defined gap in the wing as it came between the light and the eye; and this bird followed us for three days after having been fired at, though we had been sailing an average of nearly eight knots an hour.

One of the most striking examples of their endurance on the wing, however, is the fact, which we have more than once observed, that the same birds which had been unweariedly following us in the day, accompanied us throughout the whole of the succeeding night, as could be easily verified by the light of the moon.

It is a not uncommon practice with passengers to endeavour to catch these noble birds by a bait fastened to a hook and buoyed with corks. That such a cruel practice should ever be tolerated, even 'to relieve the monotony of the voyage,' is to us inconceivable, and can only be accounted for as the last resource of a brutally morbid fancy.

The albatross is essentially the scavenger of the ocean, and we doubt whether it makes any attempt to capture living fish unless when very hungry, for we have seen flying-fish rising in quantities while the albatrosses made no attempt to catch them. That the nautilus is sometimes eaten is evident, for we have taken it from the stomach; but the chief food is dead fish and other refuse. In the South Atlantic we passed the dead body of a small whale, on and around which were at least a hundred of these birds, either gorged or gorging themselves with the blubber; and guns discharged at them failed to induce many of them to take wing. We had on one occasion an opportunity of observing how rapidly these birds collect about a carcase. Like vultures or ravens, when an animal dies they discover it very speedily, and flock to the scene of the banquet. On a hot still evening in the South Atlantic a horse died, and when cast overboard next morning, the gases already formed by decomposition enabled it to float. The few albatrosses in our company immediately settled down upon it; but in less than an hour we could see through the telescope a great cloud of the birds on the sea and hovering round the unexpected prize, the almost entire absence of wind having kept us within two or three miles of the spot. It may be that the (usually) white plumage enables stragglers, far out of human ken, to see their fellows gathering in the neighbourhood of food; others again from still more remote distances may see them, and so on; until stragglers over hundreds of miles of space may be gathered to one common rendezvous.

The greater part of the year is passed by them at a distance from land; but they flock to barren and almost inaccessible rocks to breed. There the female lays her one dirty-white egg in a slight depression upon the bare

earth, the sitters being frequently so close together that it is difficult to walk without touching them. They are totally indifferent to the presence of man, and merely indicate their resent of his intrusion into their nursery by snapping at him as he passes. The parents share the labour of incubation and rearing the young, and when this is over, they all go seawards together, and silence and solitude once more reign where all had lately been clamorous and busy life.

The range of the albatross is very considerable, and it may be met with to the extreme limits of the temperate zones of both hemispheres, in the South Atlantic and North and South Pacific Oceans, both at sea and near headlands and isolated rocks. During the months of May and June in the northern, and the months of November and December in the southern hemisphere these rocks are tenanted by countless numbers of albatrosses and their smaller brown relations, known to sailors under the name of 'Mollymawks.' No one who has visited an albatross nursery will readily forget the scene. Placidly sitting upon the one precious egg is the parent, male or female as the case may be; and as far as the eye can reach over the surface, the rock is crowded with the sitters, indifferent to the presence of the human visitor. They know nothing of man's destructive nature, and they fear him not. Many of them have never seen that curious biped before, and those which have chanced to see him on his ships and to have suffered from his guns, are more likely to have then regarded him as a part of the white-sailed monster which traversed their ocean domain, than a separate creature; and fail to recognise him as he 'molests their ancient solitary reign.'

While viewing the interminable white forms thus crouching upon the earth, above wheel in graceful circles hundreds of their mates, sending congratulations in a hoarse piping voice to those beneath on the progress of the all-important business of rearing the family. Here and there sit callow uncouth nestlings; and from seawards come the parents to discharge the contents of their maws into the insatiable stomachs of the expectant young. Now and again one of the 'bread-winners' of the family swoops past the observer on its twelve feet of outspread wings, so near that he feels the shock of the divided air, and can realise the immense strength of the muscles which propel the creature, who, however, is a coward in spite of his size; for the skua gull, a bird many times smaller than himself, will often

attack him and compel him to disgorge the product of his last foraging expedition.

As soon as the albatross has reared its young, a penguin frequently takes possession of the deserted nest, and in the very cradle of a bird destined to traverse the ocean on unwearied wings lies a nestling whose wings will never develop into anything more than a pair of paddles! Great numbers of albatrosses are caught by the natives of the North Pacific coasts, who use the inflated intestines as floats for their fishing-nets, and barter the hollow wing-bones with traders for the European markets—these bones being familiar to us as pipe-stems. The large webbed feet when inflated make good tobacco-pouches. We have also seen the quills of the flight-feathers converted into floats for roach-fishing; and many a Thames angler patiently watches from his chair in the punt a feather which has probably helped to carry its former owner over the length and breadth of the Pacific.

A NOVEL LIFE-BOAT.

Mr J. Manes of Fourth Avenue, Newhaven, Connecticut, has invented a new kind of life-boat which seems to possess features worthy of notice. 'His boat consists of a hollow globe of metal or wood, ballasted at the bottom, so that it will always right itself immediately on touching the water, and can never capsize even in the roughest sea. It has compartments for water, medical stores and provisions, bull's-eyes to let in the light, a door for ingress and egress, a porthole for hoisting signals to the mast, comfortable seats all around the inside for the passengers, and a double hollow mast for supplying fresh air, and for carrying off that which has become vitiated. On the outside of the Globe boat runs a gallery, for the use of sailors in rowing, hoisting sail, discharging rockets, or steering. Of course the cases would be very rare when rowing, sailing, or steering would be required, but in case of need, all three could be easily managed.'

In such a boat—which is like a large buoy fitted with a mast—the passengers would be protected from rain and wind, and consequently to a great extent from cold. This seems to us to be a very important point, as many a shipwrecked person escapes drowning only to perish from exposure to the weather. Mr Manes suggests that a propeller might be attached to the boat to be worked by a crank turned by the passengers on the inside. It is calculated that a boat twelve feet in diameter would carry about fifty passengers, and that it could be carried on deck or hung over the stern on davits, in either of which positions it might be used as a cabin during the voyage; and further, if hung on a universal point like a compass, it would retain its equilibrium no matter what the motion of the ship might be, thus affording a safe retreat for persons subject to sea-sickness.

AWA'.

**LINES WRITTEN ON THE DEATH OF JEANIE, A
FAVOURITE CHILD, AGED SEVEN YEARS.**

THOU 'RT lyin' cauld an' still, my bonnie bonnie;
The dews o' death lie heavy on thy broo;
Thy sunny smile nae mair will thrill this bosom;
Thy sweet blue een are dark an' sightless noo!

Hushed is thy fairy tread, my bonnie bonnie;
Thy lips sae rosy red, I'll kiss nae mair.
O heavy thocht, that dims this ee wi' sadness!
O heart that fain wad break, wi' anguish sair!

I cry thy name in vain, my bonnie bonnie;
For aye thy form, thy dear-loved form I see;
O face sae fair! O locks o' golden splendour!
O guileless heart, that fondly throbb'd for me!

A dreary blank is mine, my bonnie bonnie;
Nae mair thy merry voice will cheer my ha';
An eerie stillness fills the darksome dwelling,
Since thou, my sweetest flower, wert ta'en
awa'!

The angels cam' for thee, my bonnie bonnie,
As saftly flicker'd oot life's feeble flame:
The tender Shepherd took thee to His bosom,
An' left me wi' a lanely, lanely hame!

But oh! thou 'rt wi' the blest, my bonnie bonnie,
Where pain will rend thy gentle breist nae mair;
Oh, when this weary heart lays down its sorrow,
My ain wee lassie, may I meet thee there!

JAMES SMITH.

The Conductors of CHAMBERS'S JOURNAL beg to direct the attention of CONTRIBUTORS to the following notice:

1st. All communications should be addressed to the 'Editor, 339 High Street, Edinburgh.'

2d. To insure the return of papers that may prove ineligible, postage-stamps should in every case accompany them.

3d. MANUSCRIPTS should bear the author's full *Christian* name, surname, and address, legibly written.

4th. MS. should be written on one side of the leaf only.

5th. Poetical offerings should be accompanied by an envelope, stamped and directed.

Unless Contributors comply with the above rules, the Editor cannot undertake to return ineligible papers.

Printed and Published by W. & R. CHAMBERS, 47 Paternoster Row, LONDON,
and 339 High Street, EDINBURGH.

All Rights Reserved.

*** END OF THE PROJECT GUTENBERG EBOOK CHAMBERS'S
JOURNAL OF POPULAR LITERATURE, SCIENCE, AND ART, NO.
740, MARCH 2, 1878 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.